

Project: Diarizer – SpeechBrain Conversion
Document: Environment Setup & Dependency Management

This guide outlines the standard procedure for initializing the Python virtual environment and managing project dependencies for SpeechBrain.py.

1. Environment Initialization

Navigate to the project root directory (MIT_WPU-Group-4) and initialize a fresh virtual environment:

```
python3 -m venv .venv
```

2. Environment Activation

Activate the newly created environment to ensure local dependency isolation:

```
source .venv/bin/activate
```

3. Dependency Installation

Install the required computational and audio processing libraries:

```
pip install torch torchaudio coremltools speechbrain librosa  
numpy requests
```

4. IDE Configuration (VS Code)

To resolve linting errors and ensure the IDE correctly recognizes the installed site-packages:

a. Command Palette (Cmd + Shift + P) -> "Python: Select Interpreter"

b. Select the interpreter path: `./.venv/bin/python`

5. Technical Verification

Execute a comprehensive import test to validate the environment state:

```
python -c "import torch, torchaudio, coremltools, speechbrain,  
librosa; print('Verification Successful: Environment is fully  
operational.')" 
```

Note: librosa is utilized as the primary audio loading backend to ensure high portability and compatibility without requiring additional system-level libraries.